

Lipid-Driven Organization of the Synergistic Antimicrobial Peptides PGLa and Magainin 2 on Membrane Surfaces

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ABSTRACT

The antimicrobial peptides PGLa and magainin 2 act synergistically on lipid membranes, yet their lateral organization on the membrane surface remains unclear. Here, we investigate their surface distribution using a fluorescence quenching approach¹ capable of detecting intermolecular contacts within ~1 nm. PGLa and magainin 2 were labeled with the fluorophore NBD at different amino acid positions to probe peptide–peptide proximities in membrane-bound states. Our results reveal distinct distribution patterns for both peptides that change in the presence of the synergistic partner. Based on the quenching data, we propose a distribution model describing the relative arrangement of PGLa and magainin 2 on the membrane surface. Furthermore, the observed distribution behavior strongly depends on the lipid composition, indicating the important role of lipids in peptide organization and in promoting peptide–peptide interactions. Since antibiotic peptides are well known to disturb the order of lipid fatty-acid-chains, it is likely that this disordering constitutes the driving force for the observed clustering of the peptides on the membrane surface. This effect was predicted theoretically very early in lipid research².

REFERENCES

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